

HybridRank

Intelligent Candidate Ranking · Beyond Keywords

Team **Abhijayati** · ■■■■■■■■ · To Be Victorious

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■ THE PROBLEM

Why Keyword Matching Fails Hiring

We analysed 100,000 candidate profiles. The dataset has a deliberate trap built in — thousands of Marketing Managers, HR Managers, and Accountants with AI keywords in their skills section. Any keyword ranker puts them at the top. **HybridRank doesn't.**

■ Keyword matcher

Ranks Marketing Manager #1 because their skills say 'RAG, Pinecone, Embeddings'. Misses ML engineers who built real retrieval systems but didn't use the buzzwords. Ignores that the top candidate hasn't logged in for 7 months.

■ HybridRank

Reads career history semantically. Title × domain cross-filter collapses keyword-stuffers to near-zero. Behavioral signals (last active, response rate, open-to-work) act as availability multiplier. 5 hard disqualifier rules from the JD itself.

Key stat: Only 775 of 100,000 candidates have genuine AI/ML titles (0.78%). But thousands have AI keywords sprinkled into skills with irrelevant careers. A keyword ranker fails every time.

HybridRank: Four-Layer Scoring Pipeline

Runs 100,000 candidates end-to-end in under 2 minutes on CPU. No network. No API calls. Pure Python stdlib — deterministic and reproducible.

1	2	3	4
INGEST	SCORE	PENALIZE	RANK
Load candidates.jsonl 100K profiles	4-dimension scoring per candidate	Disqualifier multipliers 5 hard rules	Top 100 + normalize to [0.20, 0.995]

■ Technical Fit	30 %	Core skills (embeddings, vector DBs, Python, ranking eval) · Proficiency × endorsements × duration · GitHub activity · Platform assessments
■ Career Quality	35 %	Title/domain relevance · Product vs consulting background · Career description semantic scan (29 regex patterns) · AI-specific YoE · Job stability
■ Behavioral	25 %	Last active date · Open-to-work · Response rate · Response speed · Interview completion · Profile completeness · Verification
■ Logistics	10 %	India location (Pune/Noida ideal) · Notice period (<30d preferred) · Work mode (hybrid/onsite) · Salary range fit

What Sets HybridRank Apart

Hard disqualifier multipliers handle every explicit red flag in the JD. These are not soft penalties — they collapse scores to near-zero for clear mismatches.

JD Red Flag	Detection	Multiplier
Keyword stuffer (wrong title + no retrieval history)	title_score<0.10 AND career_desc_relevance<0.10	×0.15
Consulting-only background (TCS/Infosys/Wipro/Accenture only)	All career companies in consulting_firms set	×0.65
Outside India, won't relocate (JD: no visa sponsorship)	country≠'India' AND willing_to_relocate=False	×0.30
Ghost candidate (inactive >180d, not open to work)	days_since(last_active)>180 AND open_to_work=False	×0.40
Recruiter black hole (response rate <10%)	recruiter_response_rate < 0.10	×0.70

Behavioral Signals as Availability Multiplier

The 23 platform signals are not just extras — recency (25%), open-to-work (20%), and response rate (20%) together determine whether a technically perfect candidate is actually hireable. A ghost candidate with a perfect skills profile still gets pushed down. This is the single biggest differentiator from keyword rankers.

■ RESULTS

Top 15 Ranked Candidates

Every candidate is a genuine AI/ML professional with product-company experience in retrieval, ranking, or recommendation systems. Zero keyword-stuffers. 99 of 100 are India-based.

#	Candidate	Score	Reasoning
1	CAND_0011687	0.9950	Senior NLP Engineer with 7.8 yrs; 5 core AI/retrieval skills; ranking/retrieval exper...
2	CAND_0002025	0.9782	Senior AI Engineer with 5.9 yrs; 6 core AI/retrieval skills; ranking/retrieval experi...
3	CAND_0064326	0.9408	Search Engineer with 7.6 yrs; 6 core AI/retrieval skills; ranking/retrieval experienc...
4	CAND_0039754	0.9332	Senior Applied Scientist with 16.2 yrs; 7 core AI/retrieval skills; ranking/retrieval...
5	CAND_0041669	0.8994	Recommendation Systems Engineer with 8.0 yrs; 6 core AI/retrieval skills; ranking/ret...
6	CAND_0088025	0.8642	Staff Machine Learning Engineer with 8.6 yrs; 7 core AI/retrieval skills; ranking/ret...
7	CAND_0046525	0.8579	Senior Machine Learning Engineer with 6.1 yrs; 4 core AI/retrieval skills; ranking/re...
8	CAND_0018499	0.8524	Senior Machine Learning Engineer with 7.2 yrs; 8 core AI/retrieval skills; ranking/re...
9	CAND_0027691	0.8245	NLP Engineer with 6.5 yrs; 5 core AI/retrieval skills; ranking/retrieval experience i...
10	CAND_0052328	0.8130	Recommendation Systems Engineer with 6.5 yrs; 4 core AI/retrieval skills; ranking/ret...
11	CAND_0036184	0.8058	Recommendation Systems Engineer with 6.0 yrs; 5 core AI/retrieval skills; ranking/ret...
12	CAND_0066999	0.8004	Recommendation Systems Engineer with 5.9 yrs; 6 core AI/retrieval skills; ranking/ret...
13	CAND_0046064	0.7972	Senior NLP Engineer with 8.9 yrs; 6 core AI/retrieval skills; ranking/retrieval exper...
14	CAND_0081846	0.7865	Lead AI Engineer with 6.7 yrs; 9 core AI/retrieval skills; ranking/retrieval experien...
15	CAND_0050454	0.7680	AI Engineer with 6.8 yrs; 4 core AI/retrieval skills; ranking/retrieval experience in...

Who made the top 10: Senior NLP Engineer @ Niramai · Senior AI Engineer @ Apple · Search Engineer @ Sarvam AI · Senior Applied Scientist @ Meta · Recommendation Systems Engineer @ CRED · Staff ML Engineer @ Yellow.ai. These are the candidates the JD is looking for.

■ DATASET INSIGHTS

What We Found in 100,000 Profiles

100K

Total Candidates

775

Genuine AI/ML titles

35.3%

Open to Work

75.1%

India-based

9,745

Consulting-only

~3,200

Keyword stuffers

~18K

Ghost candidates

99/100

Top 100 India-based

■ The keyword-stuffing trap is massive

Thousands of candidates list RAG, Pinecone, Embeddings as skills but are Marketing Managers or HR Managers. HybridRank's title × career-description cross-filter collapses their score to near-zero.

■ Behavioral signals separate paper-fit from real availability

~18% of candidates are ghost candidates — inactive for 180+ days with low response rates. Even perfect technical scores get a ×0.40 multiplier, pushing them down the list.

■ Consulting-only background is detectable and penalized

9,745 candidates spent their entire career at TCS, Infosys, Wipro, or Accenture. The JD explicitly flags this. HybridRank applies ×0.65 multiplier per the spec.

■ The score distribution validates the JD's expectations

Top candidates score 0.79–0.99. There's a clear cliff down to everyone else (<0.65). Exactly what the JD predicted: 'We're not expecting many matches in a 100K pool.'

Technical Details

Aspect	Details
Language	Python 3.11 — stdlib only (json, re, csv, datetime, multiprocessing). Zero pip installs.
Throughput	100K candidates in ~90s single-core, ~25s with multiprocessing on 4 cores.
Skill matching	frozenset lookup O(1) per candidate. No embedding computation.
Semantic scan	29 pre-compiled regex patterns on career descriptions. No model inference.
Normalization	Min-max within top-100 finalists to [0.20, 0.995]. Tie-break by candidate_id asc.
Reproducibility	Deterministic. Same output every run. No random seeds, no API calls, no network.
Run command	<code>python rank.py --candidates ./candidates.jsonl --out ./submission.csv</code>

Why no LLM inference during ranking?

The spec requires <5 minutes on CPU with no network. Instead we front-load semantic intelligence into scoring functions — 29 domain-knowledge regex patterns encode what an LLM would reason about, at rule-engine speed. Model-quality reasoning, millisecond latency.

■ SUMMARY

Why Team Abhijayati Wins

We read the JD, not just the keywords

- Parsed the actual hiring intent — product-company experience, retrieval system expertise, behavioral availability — and encoded it directly into scoring logic.
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We see through keyword stuffing

- Title × career-description cross-filter eliminates thousands of false positives. A Marketing Manager with AI skills scores 0.02, not 0.95.
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Behavioral signals are first-class citizens

- Availability is a multiplier, not an afterthought. A ghost candidate with perfect technical fit still gets pushed down if they're unreachable.
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Explainable output recruiters can trust

- Every ranked candidate has human-readable reasoning — not a black-box score.
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Production constraints respected

- Runs in <2 minutes on CPU, no network, no API keys, fully reproducible. Exactly what a real hiring platform needs at scale.
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"We'd rather see 10 great matches than 1,000 maybes." — The JD

HybridRank + Team Abhijayati delivers exactly that.